

for seasonal temperature as a substitute for ventilation. Overall, the authors found that PM_{2.5} mass higher in Ni, As, and Cr as well as Br and organic carbon significantly increased its effects on hospital admissions. For a 10 µg/m³ increase in 2-day averaged PM_{2.5}, a 1.89% (95% CI: 1.34-2.45) increase in cardiovascular disease admissions, a 2.25% (95% CI: 1.10-3.42) increase in myocardial infarction admissions, a 1.85% (95% CI: 1.19-2.51) increase in congestive heart failure admissions, a 2.74% (95% CI: 1.30-4.20) increase in diabetes admissions, and a 2.07% (95% CI: 1.20-2.95) increase in respiratory admissions were observed. The relationship between PM_{2.5} and cardiovascular admissions was significantly modified when the mass of PM_{2.5} was high in Br, Cr, Ni, and sodium ions, while mass high in As, Cr, Mn, organic carbon, Ni and sodium ions modified the myocardial infarction relationship and mass high in As, organic carbon, and sulfate ions modified the diabetes admission rates.

Emergency Hospital Admissions, All Respiratory (ICD-9 Codes 460-519)

In a single-pollutant model, the coefficient and standard error are estimated from the percent change in risk (2.07%) and 95% confidence interval (1.2% - 2.95%) for a 10 µg/m³ increase in 2-day averaged PM_{2.5} (Zanobetti et al. 2009, Table 3).

E.8 Sensitivity Analysis – At-Risk Populations

Table E-8 summarizes the PM_{2.5} health impacts functions considered by EPA to be sensitivity analyses that characterize risk experienced by certain subpopulations. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table E-8. Core Health Impact Functions for Particulate Matter Sensitivity Analyses of At-Risk Populations

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
ER Visits, Asthma	Alhanti et al.	2016	Atlanta, Dallas, St. Louis	0-4		D24HourMean	0.0025	0.0019	Log-linear	White
ER Visits, Asthma	Alhanti et al.	2016	Atlanta, Dallas, St. Louis	0-4		D24HourMean	0.0037	0.0012	Log-linear	Non-White
ER Visits, Asthma	Alhanti et al.	2016	Atlanta, Dallas, St. Louis	5-18		D24HourMean	0.0025	0.0016	Log-linear	White
ER Visits, Asthma	Alhanti et al.	2016	Atlanta, Dallas, St. Louis	5-18		D24HourMean	0.0049	0.0012	Log-linear	Non-White
Mortality, All Cause	Di et al.	2017	Nationwide	65-99	O ₃	Annual	0.0061	0.0001	Log-linear	Non-Hispanic White
Mortality, All Cause	Di et al.	2017	Nationwide	65-99	O ₃	Annual	0.0110	0.0008	Log-linear	Hispanic White
Mortality, All Cause	Di et al.	2017	Nationwide	65-99	O ₃	Annual	0.0189	0.0004	Log-linear	Black
Mortality,	Di et al.	2017	Nationwide	65-99	O ₃	Annual	0.0092	(0.0010)	Log-linear	Asian

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
All Cause										
Mortality, All Cause	Di et al.	2017	Nationwide	65-99	O ₃	Annual	0.0095	0.0019	Log-linear	Native American
Mortality, All Cause	Pope et al.	2019	Nationwide	18-99		D24HourMean	0.0095	0.0079	Log-linear	Native American, Non-Hispanic
Mortality, All Cause	Pope et al.	2019	Nationwide	18-99		D24HourMean	0.0104	0.0018	Log-linear	White, Non-Hispanic
Mortality, All Cause	Pope et al.	2019	Nationwide	18-99		D24HourMean	0.0140	0.0049	Log-linear	Black, Non-Hispanic
Mortality, All Cause	Pope et al.	2019	Nationwide	18-99		D24HourMean	0.0095	0.0079	Log-linear	Asian, Non-Hispanic
Mortality, All Cause	Pope et al.	2019	Nationwide	18-99		D24HourMean	0.0182	0.0040	Log-linear	Hispanic

E.8.1 Alhanti et al. (2016)

Alhanti et al. (2016) examined the relationship between daily PM_{2.5} concentrations and emergency room visits for asthma (ICD-9 493, 786.07) among residents of all ages in Atlanta (1993-2009), Dallas (2006-2009), and St. Louis (2001-2007). Patient-level ER visit data were obtained from hospitals in the three cities. Daily 24-hour average PM_{2.5} concentrations were estimated using data from all available monitors in the region including monitors from U.S. EPA AQS in all three cities, as well as the South Eastern Aerosol Research and Characterization (SEARCH) network and Assessment of the Spatial Aerosol Composition (ASACA) network in Atlanta. The authors ran city-specific daily time-series Poisson regression models by age group (0-4, 5-18, 19-39, 40-64, 65-99) and performed additional analysis stratified by race (White, non-White) and sex. Models controlled for temperature, day of the week, holidays, race, age, and sex.

Emergency Room Visits, Asthma

In single-pollutant models for ages 0-4, the coefficient and standard error are estimated from the three-city weighted average rate ratio (1.02) and 95% confidence interval (0.99-1.05) for White children and (1.03) and 95% confidence interval (1.01-1.05) for non-White children for a 8 µg/m³ increase in three-day moving average PM_{2.5} concentrations (Alhanti et al. 2016, Supplemental Table 4).

In single-pollutant models for ages 5-18, the coefficient and standard error are estimated from the three-city weighted average rate ratio (1.02) and 95% confidence interval (1.00-1.05) for White children and (1.04) and 95% confidence interval (1.02-1.06) for non-White children for a 8 µg/m³ increase in three-day moving average PM_{2.5} concentrations (Alhanti et al. 2016, Supplemental Table 4).

E.8.2 Di et al. (2017)

See full study description under Di et al. (2017) in Appendix E, Section E.1.1.

Mortality, All-Cause

In multi-pollutant models, the coefficient and standard error are estimated from a hazard ratio of 1.063 (95% CI: 1.060, 1.065) for White; 1.208 (95% CI: 1.199, 1.217) for Black; 1.096 (95% CI: 1.075, 1.117) for Asian; 1.116 (95% CI: 1.100, 1.133) for Hispanic; and 1.100 (95% CI: 1.060, 1.140) for Native Americans. Each odds ratio is for a 10 $\mu\text{g}/\text{m}^3$ increase in annual mean $\text{PM}_{2.5}$ exposure (Di et al. 2017, Supplementary Table S3 (GEE, By Race)).

E.8.3 Pope et al. (2019)

See full study description under Pope et al. (2019) in Appendix E, Section E.1.4.

Mortality, All-Cause

The basic CPH model is used to find the coefficient and standard error from a hazard ratio of 1.11 (95% CI: 1.07-1.15) for Non-Hispanic White; 1.20 (95% CI: 1.11-1.30) for Hispanic; 1.15 (95% CI: 1.05-1.27) for Non-Hispanic Black; and 1.10 (95% CI: 0.94-1.28) for the “Other/unknown” group, which was applied for the Non-Hispanic Native American and Non-Hispanic Asian groups. Each hazard ratio is for a 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ (Pope et al. 2019, Supplementary Table S3).

Appendix F. Core Ozone Health Impact Functions in U.S. Setup

In this Appendix, we present the core health impact functions used to estimate ozone-related adverse health effects, i.e., the functions that, as of the current release, U.S. EPA routinely uses in its regulatory analyses. Each sub-section has a table with a brief description of each health impact function and the underlying parameters. Following each table, we present a brief summary of each of the studies and any items that are unique to the study.

Note that Appendix C mathematically derives the standard types of health impact functions encountered in the epidemiological literature, such as, log-linear, logistic and linear, so we simply note here the type of functional form. And Appendix D presents a description of the sources for the incidence and prevalence data used in the health impact functions.

F.1 Short-term Mortality

Table F-1 summarizes the core health impacts functions used to estimate the relationship between ozone and mortality. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table F-1. Core Health Impact Functions for Ozone and Mortality*

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Mortality, Respiratory	Katsouyanni et al.	2009	90 U.S. Cities	0-99		D1HourMax	0.00142	0.001107	Log-linear	Warm season.
Mortality, Respiratory	Katsouyanni et al.	2009	90 U.S. Cities	0-99		D8HourMax	0.00142	0.001107	Log-linear	Warm season. 8-hour max from 1-hour max using adjustment factor of 1.13, resulting in effective beta of 0.00160.
Long-term Mortality, Respiratory	Turner et al.	2016	Nationwide	30-99	PM _{2.5} , NO ₂	Annual (D8HourMax)	0.007696	0.001176	Log-linear	Warm season.
Mortality, Respiratory	Zanobetti and Schwartz	2008	48 U.S. Cities	0-99		D8HourMax	0.000827	0.000228	Log-linear	D8HourMean approximated as D8HourMax

*Unless otherwise stated, mortality is short-term.

F.1.1 Katsouyanni et al. (2009)

Katsouyanni et al. (2009) used time series methods to examine the relationship between short-term O₃ exposures and mortality across the U.S for all ages. The study utilized mortality data from the National Center for Health Statistics (www.cdc.gov/nchs) for years 1987 through 1996, excluding accidental deaths (i.e., International Classification of Diseases [ICD]-9 800). 90 U.S. cities with population sizes